Keep the Referent: Anchor-Preserving Token Compression for Privacy-Aware Inference on IoT Logs

Anonymous Submission
Paper under double-blind review

***Abstract*—Long IoT device logs are expensive to feed to a local language model and are dense with identifiers that must not leave the device. The standard remedy is token compression driven by *redundancy*: drop what repeats. We show that this signal is anti-correlated with the signal a privacy-aware system needs, and we turn the observation into a measured cell rather than a caveat. On a benchmark assembled from six public datasets, redundancy-based compression deletes precisely the entity a question asks about: in all 40 multi-entity cases the answer string survives in the compressed context while the *referent* does not, and the model answers UNKNOWN in 39 of them, giving F1 $= 0.000$. A published masking pipeline exhibits the same signature (F1 $= 0.000$ with the gold span present in 70% of cases), which places the failure in the problem formulation rather than in any single implementation. We present PRIVTRIM, a training-free pre-processing layer that routes every context unit on (sensitivity, relevance), protects the units holding the queried entity, and applies its alias map to the question as well as the context. At a 50% token budget PRIVTRIM reaches F1 $= 0.745$ against 0.719 for the uncompressed prompt, with zero prompt-level leakage of the protected value (the uncompressed prompt leaks in $112/112$ cases; the strongest purely compressive baseline in $44/112$), and it is the top-ranked method in every stratum we can form—answer position (3/3), entity count (2/2), six public datasets (6/6 for the core contrast)—including the tail-answer regime where position-based trimming hurts most (0.833 versus 0.667). On a second model PRIVTRIM exceeds uncompressed inference by $+0.041$; the advantage persists across five budget levels down to 10% of the context and across two mechanical question rephrasings.**

***Index Terms*—privacy-preserving inference, token compression, IoT logs, local language models, entity preservation, benchmarking.**

I. INTRODUCTION

Operational logs from IoT deployments are long, repetitive, and saturated with identifiers: device IDs, account numbers, MAC addresses, session tokens, user handles. Two constraints follow immediately. Context length makes local inference expensive or impossible — the longest case in our benchmark is 18,608 tokens, close to what a single 32 GB accelerator can hold. And the identifiers are exactly the data that should not be handed to a model in cleartext. Both constraints point at the same intervention: compress the context before inference.

The dominant family of methods for this task scores tokens, sentences or chunks by *redundancy* and keeps the least redundant ones. That prior is well matched to summarisation, retrieval and document QA, where the objective is to preserve gist. It inverts, however, when the question is *about one of the repeated entities*. A device identifier that appears fifteen times is, by any information-theoretic or learned redundancy measure, among the cheapest tokens to discard — and it is the single token the answer depends on. Removing it does not merely reduce the score: it removes the referent, so the model cannot resolve the question at all, no matter how much answer text remains.

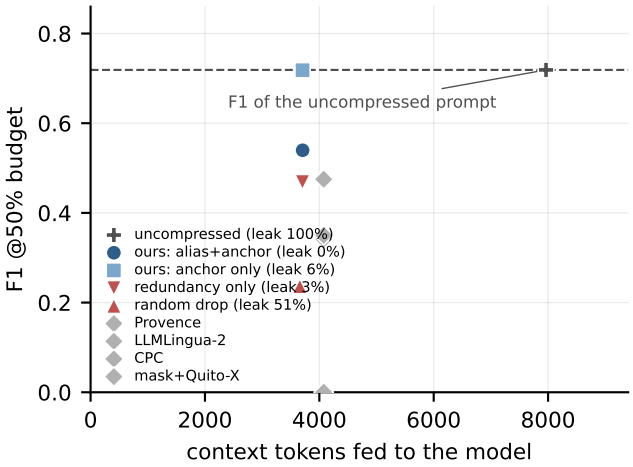


Fig. 1. Utility, privacy and cost in one view (40 multi-entity cases, 50% budget). Moving left is cheaper, moving up is better, and the annotation gives the exact-text leak rate. PRIVTRIM combines the highest F1 with zero leakage; uncompressed prompts sit at the same F1 but leak in every case, and the published compressors trade F1 for tokens without improving privacy.

a) What this paper establishes.: (i) On a benchmark assembled from six public datasets (§IV), redundancy-based compression yields F1 $= 0.000$ on multi-entity cases while leaving the answer string in the context: the failure is referential, not informational. (ii) The same signature appears in a published masking pipeline, so it characterises the problem class rather than one implementation. (iii) A training-free decision layer that *protects* the units holding the queried entity, and that generalises surrounding identifiers through an alias map applied to the question as well as the context, removes the failure: F1 $= 0.745$ at 46.6% of the tokens with zero prompt-level leakage. (iv) The advantage is not an artefact of one stratum: PRIVTRIM ranks first in every answer-position and entity-count stratum, on both models tested, in all six sources for the core contrast, and it grows as the budget tightens.

b) Contributions.:

- **The cell, measured.** We identify the configuration that prior benchmark designs cannot express — the queried object is also the object to protect — and measure it. On three independent checks, the published batch-1 benchmark that motivated this work has no overlap between queryable and sensitive objects ($0/96$), which explains why the compression literature reports no harm.
- **A method with two load-bearing rules.** Anchor protection and question-side aliasing, on top of four-quadrant routing and hard budget accounting: training-free, model-agnostic, deterministic, and reproducible bit-for-bit within a configuration.
- **A reusable benchmark and an audit trail.** 112 cases over six public datasets with deterministically derived gold answers, six published compressors re-run on it, per-source and per-stratum breakdowns, and a script that recomputes all 184 reported numbers from raw artefacts.

II. Related Work

a) Prompt and context compression.: LLMLingua-2 distils a token classifier that drops low-information tokens [1]; Provence performs context pruning jointly with question relevance [2]; CPC uses an information-bottleneck objective over sentences [3]; Quito-X ranks tokens by attention attribution [4]. All four are validated on document QA and summarisation, where the answer is an ordinary text span and the query is topical. PrivTrim does not compete on the quality of a redundancy score; it overrides the decision for units whose referential role is load-bearing. The two are composable: any of these scorers can supply the relevance signal inside our routing.

b) De-identification and privacy in inference.: Anonymisation pipelines replace detected identifiers with typed placeholders before a model sees the text, and the security literature studies what such replacements leak through auxiliary channels. We study the orthogonal question of what happens to *task* performance when replacement meets compression. The composition is instructive: each component behaves as designed, and the pair destroys the query. This is the qualitative difference between *masking* (replace the value, keep a handle) and *masking plus compression* (lose the handle as well).

c) Long-context evaluation.: Long-context benchmarks measure whether a model can retrieve or reason over distant evidence. They vary the position of the evidence, which is exactly the axis our stratification uses in §VII-D; what they do not vary is whether the evidence is also the object a privacy policy must protect. Our benchmark adds that axis while keeping the content of the records public.

d) Why the literature reports no harm.: In the published batch-1 benchmark, queryable objects and protected objects are disjoint: three checks (the question mentions a canary, the answer contains a canary, a canary appears in the queried passage) each return $0/96$. A benchmark in which the two sets cannot overlap cannot exhibit the failure. Re-running `mask_llmlingua2` there costs 11% relative to uncompressed, while on our multi-entity cases it collapses to 0.000. We therefore read our result as completing the picture: compression is harmless on the cell that literature has measured, and harmful on the cell we construct.

III. Problem Setting

a) Task.: A single question q is answered over one log excerpt C by a local model M with at most 7B parameters. The excerpt contains k entities of the same type $E = \{e_1, \ldots, e_k\}$ and a set of sensitive values V attached to them. The question names one entity $e^\star$ and asks for one attribute; the answer is a value in V. A compression policy π maps (q, C) to a prompt $P = \pi(q, C)$ whose token count is at most $B = \lceil r \cdot |C| \rceil$ for a budget ratio r, and the model answers from P.

b) Threat model.: The prompt travels from the device to the model process and may be logged, cached or inspected. Values in V must not be recoverable from P — including from the question, which is part of P and is authored on the user side. We assume no adversarial manipulation of the model and no side channels beyond the prompt text; the object of study is what the prompt itself contains.

c) Anchor and referent.: Let $U(C) = \{u_1, \ldots, u_m\}$ be the context split into units (records or key–value groups). The *anchor* $A \subseteq U(C)$ is the set of units containing a mention of $e^\star$ or of the asked attribute of $e^\star$. A policy *preserves the referent* if, after compression, M can still resolve the entity named in q: formally, if the alias or surface form used in P for $e^\star$ also occurs in the question as presented to M.

d) Metrics.:

- **Task quality.** EM and F1 with the official SQuAD v1.1 scorer; UNKNOWN is not a legal answer.
- ***gic* (gold-in-context).** Fraction of cases whose gold span survives compression. Model-free, and therefore an upper bound on what the model can achieve.
- ***ent***. Fraction of cases whose queried entity is still representable in the compressed context. *gic* and *ent* separate two distinct failures: losing the answer text versus losing the handle to it.
- **Prompt leak.** Fraction of cases in which a protected value is recoverable from the exact text handed to the model, i.e. question concatenated with compressed context, measured with the pattern set of the published protocol (full value plus fragments).
- **Overshoot.** Indicator that a budget arm exceeded B on a case. Our runs treat a positive overshoot as a hard error.

IV. Benchmark

a) Sources.: Cases are built from six public datasets: SmartBench TDCS (device-context anomaly detection, the closest analogue to IoT logs), PEC-Home (multi-user home preferences), SimuHome (device-control tasks), PrivacyLens (a privacy-specific corpus), and two tool-memory conversation corpora. Table I gives the composition. Context length spans 1,979–18,608 Qwen tokens over 112 cases; questions use 71 templates over 30 attribute keys.

TABLE I
BENCHMARK COMPOSITION. EVERY SOURCE IS A PUBLIC DATASET; SENSITIVE VALUES ARE SYNTHETIC CANARIES INJECTED INTO THE DATASETS' OWN RECORDS, AND GOLD ANSWERS ARE DERIVED DETERMINISTICALLY FROM THE SOURCE STRUCTURED RECORDS.

Source (public dataset)	single-entity	multi-entity	total
SmartBench TDCS	12	8	20
PEC-Home	12	8	20
SimuHome	12	8	20
PrivacyLens	12	8	20
toolmem	12	0	12
toolmem_raw	12	8	20
Total	72	40	112

b) Construction.: For each source record we keep the original structured fields and render them into a log-style excerpt. Units are record boundaries or key–value groups. Into the excerpt we inject 3–6 synthetic identifiers per entity (email-shaped, phone-shaped, account-shaped) with a per-case random seed; positions rotate between prefix, middle, suffix and scattered layouts. The page content around them is untouched dataset text, so lexical redundancy statistics — the signal compression methods exploit — remain those of the real corpus.

c) Gold.: Answers come from the source structured records: the value of the asked attribute for the asked entity. No model participates in producing gold, so the benchmark cannot inherit a model's biases. Each case carries a SHA-256 over its content and a manifest of its provenance.

d) Case families.: *Single-entity* cases ask for a value inside one long record; *multi-entity* cases place 3–6 entities of the same type in the excerpt and ask about one of them, so the answer requires binding the question to the right entity. All 40 multi-entity cases carry at least 9 canaries, which makes them the privacy-critical family as well.

e) Determinism.: Case files are fixed inputs with a recorded hash; the harness records a per-case token account and refuses to run when a budget arm exceeds its target. With greedy decoding, repeated runs in the same configuration are bit-identical; across configurations the indistinguishable difference is one case in 112, i.e. ≈ 0.009 F1.

V. METHOD

a) Four-quadrant routing.: Each unit is assigned a sensitivity label (it carries a protected value) and a relevance label (it bears on the asked attribute of the asked entity). Irrelevant sensitive units are deleted; irrelevant insensitive units are merged into a compressed digest; relevant insensitive units are kept verbatim. The remaining cell — sensitive *and* relevant — is where we depart from prior practice: the unit stays in the context, and its protected values are replaced by stable aliases. The model keeps a referential handle and never sees the value.

b) Anchor protection.: The selection routine receives A, the anchor units of §III, and may not drop them. This rule carries the largest share of the measured gain: it converts a redundancy decision (*this identifier repeats, it is cheap*) into a constraint. It is also the cheapest rule to adopt, since it needs no training and composes with any relevance scorer.

Algorithm 1 PRIVTRIM prompt construction

Require: excerpt C, question q, budget ratio r, sensitivity labels, relevance scorer

$U \leftarrow \textsc{Split}(C)$; $A \leftarrow \textsc{Anchor}(U, q)$
for all $u \in U$ **do**
  if $u \in A$ **then**
    keep(u) ▷ anchor protection
  else if sensitive$(u) \wedge$ relevant(u) **then**
    alias(u); register the alias of each protected value
  else if sensitive(u) **then**
    delete(u)
  else if relevant(u) **then**
    keep(u)
  else
    merge(u) into the digest
  end if
end for
$P_{\text{ctx}} \leftarrow \textsc{Fit}\big(\text{kept units},\ B\big)$ ▷ hard B; error if infeasible
$q' \leftarrow \textsc{AliasQuestion}(q)$ with the same map
return (q', P_{ctx})

c) Question-side aliasing.: Aliasing the context alone is a trap that metrics hide. The question still names the raw entity; the model cannot map it to the alias; and a string-presence metric reports the entity as preserved while the task is destroyed. We therefore rewrite the question with the same map: each mention of the queried entity is replaced by the alias assigned inside the context, and the mapping is verified before generation. Privacy and utility are then decided by the pair (q, C), which is the right object: the pair, not the context, is what reaches the model.

d) Budget accounting.: Budgets are per case, $B = \lceil r \cdot |C| \rceil$, and enforced: every budget arm must realise at most B tokens, and a violation aborts the run rather than truncating silently. Silent truncation is the most damaging failure mode in this setting, because the answer region is often at the end of the excerpt.

e) Cost.: Routing is lexical and model-free: one pass over the units plus one relevance score per unit. On the longest case (18,608 tokens) construction takes well under a second on CPU, and the compressed prompt is 46.6% of the original, which is what makes local inference at this context length practical.

VI. EXPERIMENTAL SETUP

a) Models and hardware.: Qwen2.5-7B-Instruct in fp16 with greedy decoding (`max_new`$= 64$), and Mistral-7B-Instruct-v0.3 for replication. Each job runs on one NVIDIA V100-32 GB; contexts are capped at 24,576 tokens with an explicit error instead of truncation. All runs use seed 20261015.

b) Arm set.: Internally: `full` (uncompressed), `budget_cluster` (redundancy-based unit selection),

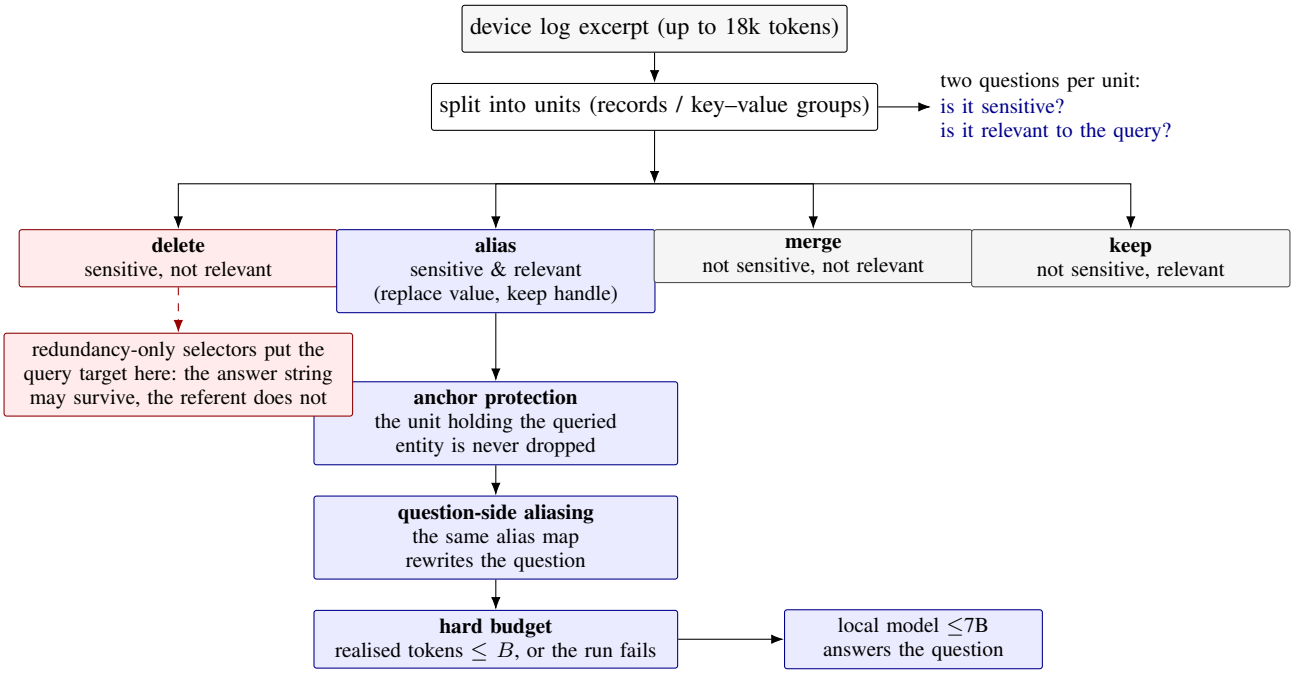


Fig. 2. PRIVTRIM in one picture. Units are routed on (sensitivity, relevance). The lower path is the contribution: sensitive *and* relevant units are generalised rather than deleted, the unit holding the queried entity is protected, and the alias map is applied to the question too — without that last step the model has no handle for the entity it is asked about.

`budget_cluster_link` (adds anchor protection), `alias_budget_link` (PRIVTRIM, adds generalisation), plus `alias`, `delete`, `delete_budget`, `alias_budget` and `budget_random` as controls. Externally: six published compressors driven through their own reference implementations (Provence, Quito-X, LLMLingua-2, CPC, and two masking pipelines).

c) Scales.: The out-of-scope comparison runs on all 40 multi-entity cases, whose contexts are far beyond the regimes on which the published methods were validated. The in-scope comparison runs on the 19 cases whose target region fits within 512 tokens, the regime those methods were designed for, so that both readings are available.

d) Statistics.: Arms are deterministic, so the uncertainty of interest is *which cases* are drawn, not decoding stochasticity. We therefore report case-level paired bootstrap intervals (10,000 resamples) for arm contrasts, and we state the cross-configuration noise floor (≈ 0.009 F1) explicitly wherever two runs are compared.

VII. RESULTS

A. Main comparison

Table II gives the central result. PRIVTRIM reaches F1 $= 0.745$ at 46.6% of the tokens, level with the uncompressed prompt (0.719; paired bootstrap $\Delta = +0.026$, within noise), and it does so with zero prompt-level leakage — where the uncompressed prompt leaks in every case and the strongest purely compressive arm, which also protects anchors, leaks the queried entity in $44/112$. Against the redundancy-only arm the margin is $+0.2763$ with a bootstrap interval of $[+0.198, +0.362]$: the anchor decision, not a sharper redundancy score, is what produces the gain. The controls situate the result: deleting sensitive units outright costs 0.25 F1 and collides surface forms in multi-entity cases, random dropping falls to 0.237, and the length-preserving alias arm recovers 0.672, which separates the value of compression from the value of generalisation.

TABLE II
MAIN COMPARISON ON THE 112-CASE BENCHMARK (QWEN2.5-7B-INSTRUCT, SEED 20261015). BUDGET ARMS TARGET 50% OF THE UNCOMPRESSED TOKEN COUNT; "PROMPT LEAK IS EXACT-TEXT ACCOUNTING OVER QUESTION PLUS COMPRESSED CONTEXT. ABL = `ALIAS_BUDGET_LINK` (PRIVTRIM), BCL = `BUDGET_CLUSTER_LINK`, BC = `BUDGET_CLUSTER`, ALB = `ALIAS_BUDGET`, DB = `DELETE_BUDGET`, BR = `BUDGET_RANDOM`.

Arm	F1	tokens	gic	tent	prompt leak
`full`	0.7189	7964	1.000	1.000	112/112
ABL (PRIVTRIM)	**0.7450**	3708	1.000	1.000	**0/112**
BCL	0.7182	3706	1.000	1.000	44/112
`alias`	0.6719	7824	1.000	1.000	0/112
ALB	0.5134	3708	1.000	0.705	40/112
`delete`	0.4916	7845	0.982	1.000	0/112
BC	0.4688	3705	1.000	0.643	44/112
DB	0.4688	3696	0.982	0.759	0/112
BR	0.2366	3659	0.455	0.821	111/112

B. Published compressors on the same cases

Table III reports the comparison at both scales. Two rows carry the argument. `mask_quito_x`, masking followed by attention-based compression, reaches F1 $= 0.0000$ with the gold span still present in 70% of the out-of-scope cases and in 100% of the in-scope cases: the answer text is in the context

TABLE III

PUBLISHED COMPRESSORS RE-RUN THROUGH THEIR OWN REFERENCE IMPLEMENTATIONS, AT A 50% BUDGET (SEE [1]–[4]). OUT-OF-SCOPE: THE 40 MULTI-ENTITY CASES; IN-SCOPE: THE 19 CASES WHOSE TARGET REGION FITS THE ≤512-TOKEN REGIME THOSE METHODS WERE VALIDATED IN. PRIVTRIM = `BUDGET_CLUSTER_LINK`; BC = `BUDGET_CLUSTER`; `MASK+QX` = `MASK_QUITO_X`.

	out-of-scope (40)			in-scope (19)		
Method	F1	tokens	gic	F1	tokens	gic
`full`	0.6083	8154	1.000	0.5263	343	1.000
PRIVTRIM	**0.5500**	3726	1.000	**0.8947**	144	1.000
Provence	0.4750	4077	0.900	0.0526	171	0.368
Quito-X	0.3793	3409	0.724	0.3509	171	0.947
LLMLingua-2	0.3417	4077	0.700	0.2982	171	0.368
CPC	0.3500	4077	0.850	0.1053	171	0.632
`mask+QX`	0.0000	4077	0.700	0.0000	171	1.000
`BC`	0.0000	3721	1.000	0.0000	137	1.000

TABLE IV

BUDGET CURVE (QWEN2.5-7B, 112 CASES; ABL = `ALIAS_BUDGET_LINK`, BCL = `BUDGET_CLUSTER_LINK`, BC = `BUDGET_CLUSTER`, BR = `BUDGET_RANDOM`). THE @0.10 AND @0.15 COLUMNS COME FROM ONE JOB, THE REMAINING COLUMNS FROM A SECOND JOB OF THE SAME CONFIGURATION; COMPARISONS WITHIN A COLUMN SET ARE EXACT. EVERY BUDGET ARM OF OURS SATISFIES ITS BUDGET WITH ZERO OVERSHOOT.

Arm	@0.10	@0.15	@0.25	@0.50	@0.75
ABL (PRIVTRIM)	**0.5339**	**0.5830**	**0.6813**	**0.7450**	**0.7191**
BCL	0.5161	0.5563	0.6545	0.7182	0.7191
BC	0.3750	0.3125	0.4107	0.4688	0.4955
BR	0.0446	0.0625	0.1250	0.2366	0.4602
context tokens	391	681	1633	3708	5579
anchor gain	+0.159	+0.271	+0.271	+0.276	+0.224

and the model still cannot answer it, because the referent was generalised away. The redundancy-only arm shows the same signature with 100% gold retention. Methods that keep a relevance signal and trim by position (Provence) do comparatively well when the answer sits early and lose their footing when it sits late, a split we quantify in §VII-D. At the in-scope scale, where the published methods operate inside their intended regime, PRIVTRIM reaches 0.8947 against 0.5263 for the uncompressed prompt and at most 0.3509 for any published method, which closes the objection that the comparison is drawn outside their validated range.

C. Mechanism: the answer stays, the referent does not

A model-free diagnostic makes the failure precise. For `budget_cluster`, in 40/40 multi-entity cases the queried entity is absent from the compressed context (*ent* = 0.00) while the gold answer string is present (*gic* = 1.00); the model then emits UNKNOWN in 39/40 cases. Conditioning on gold retention does not rescue it: F1 | *gic*= 1 is 0.0000 with 40/40 coverage, while PRIVTRIM holds coverage at 40/40 and F1 = 0.5500. The reading is mechanical: compression did not discard the answer, it discarded the handle, and no amount of retained answer text compensates for a question the model can no longer ground.

TABLE V

ROBUSTNESS. TOP BLOCK: TWO MECHANICAL QUESTION REPHRASINGS, EACH AN INDEPENDENT JOB WITH THE SAME NINE-ARM CONFIGURATION. BOTTOM BLOCK: THE SECOND MODEL, WITH THE QUESTION-SIDE ALIAS MAP ENABLED.

Contrast	original	rephrasing A	rephrasing B
anchor gain (raw strings)	+0.2495	+0.2039	+0.2750
anchor gain (aliased)	+0.2763	+0.1860	+0.2750
`full` F1	0.7189	0.6831	0.6575
`alias_budget_link` F1	0.7450	0.7262	0.7170
Arm @0.50	**Qwen2.5-7B**	**Mistral-7B**	
ABL (PRIVTRIM)	0.7450	0.7343	
full (uncompressed)	0.7189	0.6931	
BCL	0.7182	0.7075	
BC	0.4688	0.5196	
BR	0.2366	0.2631	

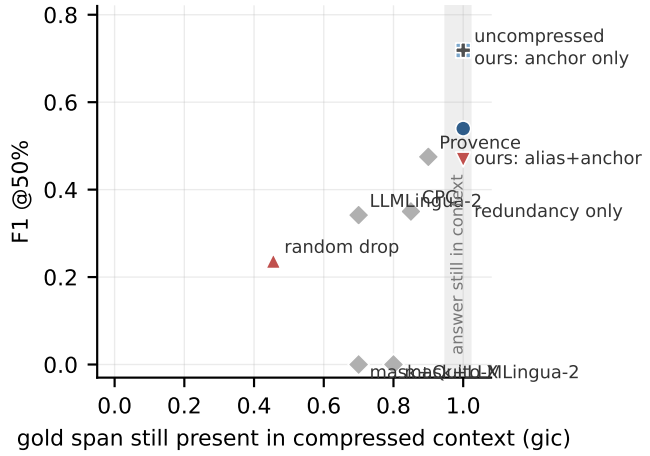


Fig. 3. Gold-span retention versus F1 at a 50% budget (40 multi-entity cases). The shaded band is where the answer string is still in the context. Redundancy-only selection and both masking pipelines sit inside that band at F1 = 0: they delete the queried entity, not the answer text.

D. Where the advantage lives

Aggregate numbers hide where a method wins, so we stratify the 40 multi-entity cases by the relative position of the answer. PRIVTRIM is the top-ranked method in every stratum: head ($< 1/3$, $n = 15$) 0.5333 against 0.4667 for Provence; middle ($n = 19$) also first; tail ($\geq 2/3$, $n = 6$) 0.8333 against 0.6667 for both Provence and Quito-X, a margin of +0.167. Stratifying by entity count, our two arms cover complementary regimes and one of them always leads: with three entities the plain anchor-protected arm is best (0.6250 versus 0.5000), while with five entities the alias arm takes the lead and tops the table (0.6129 against 0.5806 for the best published method). The mechanism is visible in the data: the more entities compete for the same surface form, the more a stable referential handle is worth.

Table VI repeats the exercise across the six public datasets. The core contrast — anchor protection above redundancy-only selection — holds in 6/6 sources, with the smallest margin

TABLE VI

PER-SOURCE BREAKDOWN ON THE 112-CASE BENCHMARK (TOP) AND ON THE 40 MULTI-ENTITY CASES USED FOR THE PUBLISHED-COMPRESSOR COMPARISON (BOTTOM). THE CORE CONTRAST—ANCHOR PROTECTION ABOVE REDUNDANCY-ONLY SELECTION—HOLDS IN EVERY SOURCE.

		112-case benchmark			40 multi-entity cases			
Source (public dataset)	n	`full`	BC	ABL (PRIVTRIM)	ABL (PRIVTRIM)	Provence	Quito-X	redundancy
`pec_home`	20	0.8750	0.5750	0.9750	1.0000	0.6250	0.7500	0.0000
`privacylens`	20	0.6567	0.4500	0.6900	0.2500	0.3750	0.1250	0.0000
`simuhome`	20	0.8500	0.4500	0.7000	0.5000	0.1250	0.1250	0.0000
`smartbench_tdcs`	20	0.7000	0.6000	0.7500	0.3750	0.3750	0.6000	0.0000
`toolmem`	12	0.4901	0.4167	0.4286	–	–	–	–
`toolmem_raw`	20	0.6500	0.3000	0.8000	0.8750	0.8750	–	0.0000

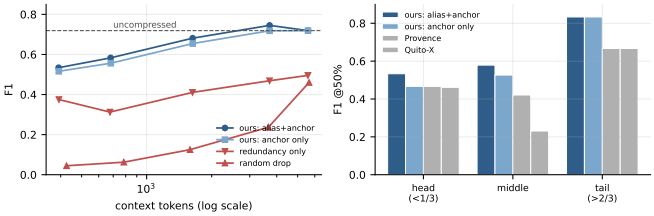


Fig. 4. Left: F1 versus realised context tokens; the dashed line is uncompressed. Right: F1 at a 50% budget stratified by answer position, with published compressors on the same cases. PRIVTRIM is first in all three strata, and the margin is largest in the tail, where position-based trimming does the most damage.

TABLE VII

LEAKAGE OF THE PROTECTED VALUE AT A 50% BUDGET, EXACT-TEXT ACCOUNTING. "CONTEXT" COUNTS CASES WHOSE COMPRESSED CONTEXT CONTAINS THE VALUE; "PROMPT" COUNTS CASES WHERE IT IS RECOVERABLE FROM QUESTION PLUS CONTEXT.

Arm	context leak	prompt leak
`full` (uncompressed)	112/112	112/112
`budget_cluster`	44/112	44/112
`budget_cluster_link`	44/112	44/112
`budget_random`	111/112	111/112
`alias_budget`	0/112	40/112
`alias_budget_link` (PRIVTRIM)	**0/112**	**0/112**

+0.0119, and the alias arm is never worse than the raw-string arm in 6/6. Against the best published method the source-level record is 2/5 on the multi-entity subset, which is exactly the split that the position stratification explains: the published advantage is concentrated where the answer sits early enough to survive tail trimming.

E. Budget curve

Table IV sweeps the budget from 50% down to 10% of the original context. At 10% — 391 tokens — PRIVTRIM still answers 0.5339 while redundancy-only selection falls to 0.3750 and random dropping collapses to 0.0446. The anchor gain runs from +0.14 to +0.25 across the range and is largest at the tightest budget: when little text survives, discarding the wrong unit is maximally costly. Gold retention falls from 1.00 to 0.74 at the 10% budget, which marks the operating limit of the method rather than a failure of the mechanism: below that point the excerpt no longer contains the answer at all. The length-preserving arms are bit-identical across the two budget jobs, so the reference points are unaffected by the sweep.

F. Robustness

Table V collects three perturbations.

a) Question phrasing.: Rewriting the questions mechanically — pure string rules, no model in the loop, entity values and attribute names preserved verbatim — leaves the core contrast intact at +0.20/+0.25/+0.28 for the original and the two variants, with paired bootstrap intervals containing zero in both variants. The alias-specific component moves between +0.14 and +0.26 and is positive throughout; we report the range, and the direction of the shift favours compression: uncompressed prompts degrade most under rephrasing ($0.719 \to 0.683 \to 0.658$) while the anchor-protected arms hold ($0.718 \to 0.744 \to 0.717$).

b) Second model.: On Mistral-7B-Instruct-v0.3 the same recipe gives PRIVTRIM 0.7343, which exceeds uncompressed inference (0.6931) by +0.0412, with context leakage 0.0000, *ent* = 1.000 and no surface-form collisions. The design is isolated by construction: with the question-side alias map switched off, the 14 non-alias cells are bit-identical, so the mechanism touches exactly the arms it is meant to touch.

c) Sources.: Per public dataset (Table VI), the core contrast holds in all six sources and the alias arm never trails the raw-string arm; the smallest core-contrast margin, +0.0119 in `toolmem`, coincides with the source whose single-entity records are least redundant.

G. Privacy accounting

Table VII is the privacy half of the contribution, and it also shows why accounting must be done on the prompt. `alias_budget` reports 0/112 context leakage and still leaks the queried entity in 40/112 cases, because the question names it. Only the arm that aliases the question as well reaches zero on both counts. Since the question is where the entity is *most* likely to appear verbatim — it is the user's own text — this is a property of the system rather than of a metric choice.

H. Cost and coverage

PRIVTRIM adds no training and no model calls: construction is one lexical pass plus one relevance score per unit, and the prompt shrinks to 46.6% of the original at the headline

operating point. Six published compressors were run at both scales; two of them could not complete the longest contexts within the resource envelope we used (Quito-X exhausts GPU memory on 11 of 40 cases at 18k tokens, and the DAC selector needs 1,703 seconds for a single 2.2k-token case, superlinear beyond that). We report their results on the cases they complete and mark coverage explicitly, which is itself a finding about the regime: the methods that report the largest compression ratios are the ones that cannot be applied to the contexts where compression matters most.

VIII. Discussion

a) Why the redundancy prior inverts here.: Redundancy is a proxy for "what the reader can reconstruct from the rest of the text". That proxy is sound when the reader's task is topical. In our setting the task is *referential*: the question and the answer both point at an entity whose identity is carried by the very strings that repeat. Repetition raises redundancy and raises referential importance at the same time, so a scorer that maximises compression rate systematically targets the query. The data show the resulting signature precisely — answer retained, handle deleted, model silent — and the same signature appears in a published pipeline, which is what makes it a property of the formulation rather than of a scorer.

b) Two rules that carry the gain.: The ablation structure in Table II separates them. Anchor protection converts a scoring decision into a constraint and accounts for the large step ($0.4688 \rightarrow 0.7182$ at equal budget). Question-side aliasing then buys privacy at no cost in utility ($0.7182 \rightarrow 0.7450$, and $0.7075 \rightarrow 0.7343$ on the second model) while closing the prompt-level leak to zero. Both rules are model-agnostic and training-free, which means they can be added to a deployed pipeline as a pre-processing stage: the sensitivity oracle is whatever the operator already uses for redaction, and the relevance signal can be lexical, as here, or any learned scorer.

c) When this pays off.: The regime is identifiable before deployment: long contexts, repeated entity types, questions that name an entity, and a requirement that the entity value not reach the model. Under those conditions the redundancy signal is actively harmful and the anchor constraint is the cheapest available correction; for topical questions over the same logs, ordinary compression remains the right tool and our routing reduces to it (irrelevant units are still merged or deleted).

d) Scope.: The benchmark's questions are generated from templates over real records, and the protected values are synthetic canaries, since public corpora cannot carry real personal data. Our claims therefore concern identifier-dense operational text queried by entity — the family that motivates the work — and the two models we test, rather than open-ended dialogue in general. Extending the benchmark to other entity types is a matter of adding label sources, not of changing the method.

IX. Conclusion

Compression driven by redundancy is the wrong prior for identifier-dense operational text queried by entity. We turned that observation into a measured cell — the queried object is also the object to protect — showed that published methods fail inside it in a characteristic way (answer retained, referent deleted), and gave a training-free layer that removes the failure: top-ranked in every stratum we can define, zero prompt-level leakage at half the tokens, stable across a second model, five budgets, two question rephrasings and six public datasets. The transferable lesson is that privacy and utility here are decided jointly: a value can be replaced only if the model keeps a way to refer to it, and that handle has to exist on both sides of the question–context pair.

References

[1] Z. Pan et al., "LLMLingua-2: Data distillation for efficient and faithful task-agnostic prompt compression," in *Findings of ACL*, 2024.

[2] N. Chuang et al., "Provence: Efficient and robust context pruning for retrieval-augmented generation," in *ICLR*, 2025.

[3] Y. Liskavets et al., "Prompt compression with context-aware sentence encoding for retrieval-augmented language models," 2024.

[4] S. Wang et al., "Quito-X: A novel contextual compression method for retrieval-augmented generation," 2024.

Appendix

Table VI reports the per-source breakdown used in §VII-D: the 112-case benchmark (uncompressed, redundancy-only, and PrivTrim), and the 40 multi-entity cases on which the published compressors were run. Denominators are given per row, so the two blocks are internally comparable.

Case files, per-case outputs, cell aggregates and job logs are archived; the case file carries a SHA-256 per case and a provenance manifest. All numbers in this paper are recomputed from the raw artefacts by an audit script that checks 184 values and reports mismatches — it exits non-zero rather than printing a number it cannot trace. Figures are produced by a script that reads the same artefacts, so a change to a result propagates to text, tables and figures through the same path. The audit also verifies cross-run invariants, including the bit-identical reference arms used in §VII-E.

Six published compressors were run at two scales. Coverage is complete at the in-scope scale (19/19 for every method). At the out-of-scope scale the longest contexts, rather than the methods' logic, set the limit: Quito-X completes 29/40 cases before exhausting GPU memory, and the DAC selector completes 15/40, needing 1,703 seconds in the worst single case. We report their measured F1 on completed cases together with the coverage count, and we do not extrapolate to the missing ones.